

Exoskeleton outer layer, or cuticle, contains shiny, waterproofing wax



Discarded, hollow cuticle, complete with body hairs



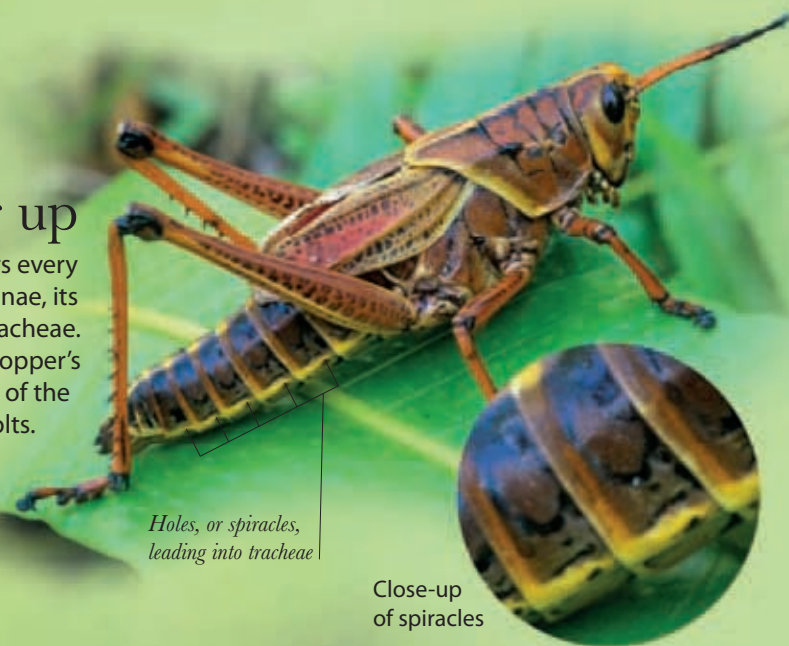
New coat

Hanging upside down by a thread of silk, this golden orb-web spider has just shed, or molted, the outer “cuticle” layer of its exoskeleton. Molting is a complex operation, and it begins when a spider’s old cuticle separates from the new one growing beneath. The old cuticle splits along the back, and the spider climbs out, extracting its legs like someone pulling their fingers from a glove. Some arthropods molt throughout their lives, but most insects stop molting once they have become adults with fully working wings.

Spider expands as new cuticle dries and hardens

Cover up

A grasshopper’s exoskeleton covers every part of its body, including its antennae, its eyes, and even its breathing tubes, or tracheae. These tubes reach deep inside the grasshopper’s body. Their linings are shed with the rest of the cuticle, each time the grasshopper molts.



Holes, or spiracles, leading into tracheae

Close-up of spiracles



Underwater giant

The Japanese spider crab is the world’s largest arthropod, with a maximum leg span of nearly 13 ft (4 m). Like other crustaceans, it has a calcified exoskeleton—one that is reinforced with the mineral calcium carbonate. It has a shield, called a carapace, which protects its head, and powerful claws that can crack open shells. On land, without water to support its weight, the spider crab can barely move.



Male’s abdomen clasps female

Female’s abdomen bends forward to make contact with male

Elegant embrace

Locked together in a typical mating wheel position, these damselflies show how flexible exoskeletons can be. Their long abdomens are made up of many jointed sections that enable the male to arch its body and clasp the female behind the head during mating.